

Vedic Mathematics Sutras :

Notes by : Dr. Snehal K Joshi

Why Learn Vedic Mathematics?

- Improves mental calculation speed
- Reduces dependency on calculators
- Builds logical and analytical thinking

1 Nikhilam Navatashcaramam Dashatah

Meaning: *All from 9 and the last from 10*

Concept:

This sutra is used for fast subtraction and multiplication when numbers are close to 10, 100, 1000, etc.

Application:

- Subtract each digit from 9 (and last digit from 10)
- Useful for mental arithmetic and quick calculations

Example:

$$1000 - 357 = 643$$

$$(9-3=6, 9-5=4, 10-7=3)$$

2 Ekadhikena Purvena

Meaning: *By one more than the previous one*

Concept:

This sutra is used for squaring numbers ending in 5 and special multiplication cases.

Application:

- Multiply the first digit by one more than itself
- Append 25 at the end

Example:

$$35^2 \rightarrow (3 \times 4) = 12 \rightarrow \text{Answer} = 1225$$

3 Udharan

Meaning: *The extraction*

Concept:

This sutra is used for extracting roots such as square root and cube root.

Application:

- Helps simplify complex root calculations
- Enables faster mental extraction of roots

Example:

$$\sqrt{144} = 12 \text{ (quick identification using pattern)}$$

4 Paraavartya

Meaning: *Transposition and cancellation*

Concept:

This sutra is used for division and algebraic simplification.

Application:

- Transpose terms and cancel common factors
- Useful in solving equations quickly

Example:

Used in dividing large numbers and solving polynomial equations.

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5. Shunyam Saamyasa-muccaye

Meaning: When the totals on both sides are equal, their difference is zero.

Explanation

This sutra is used when two expressions have the same sum. If both sides are equal, we can directly equate their remaining parts to find the unknown value.

Example

$$2x + 5 = x + 12$$

$$2x - x = 12 - 5$$

$$x = 7$$

6. Anurupyena

Meaning: Proportionately

Explanation

This sutra is useful in ratio and proportion problems. It helps in solving problems where quantities increase or decrease in the same proportion.

Example

If 4 pens cost ₹40

Then 1 pen = ₹10

So 6 pens = ₹60

7. Sankalana–Vyavakalan-abhyam

Meaning: By addition and subtraction

Explanation

This sutra is used to solve equations by adding or subtracting values from both sides to isolate the unknown variable.

Example

$$x + 9 = 17$$

$$x = 17 - 9$$

$$x = 8$$

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8. Purana-puran-abhyam

Meaning: By completion or non-completion

Explanation

This sutra helps in completing expressions to make perfect squares or simpler forms. It is very useful in algebraic factorization.

Example

$$\begin{aligned} x^2 + 4x \\ = (x + 2)^2 - 4 \end{aligned}$$

9. Chalana–Kalana

Meaning: By shifting or adjustment

Explanation

This sutra allows us to shift numbers to make calculations easier by using nearby round numbers.

Example

$$\begin{aligned} 99 \times 6 \\ = (100 - 1) \times 6 \\ = 600 - 6 = 594 \end{aligned}$$

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10. Yavadunam

Meaning: Whatever is the deficiency

Explanation

Used for squaring numbers close to 10, 100, 1000 etc. It uses the deficiency from the base for fast calculation.

Example

$$97^2$$

$$\text{Base} = 100$$

$$\text{Deficiency} = 3$$

$$\text{Left: } 97 - 3 = 94$$

$$\text{Right: } 3^2 = 09$$

$$\text{Answer} = 9409$$

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11. Vyastisamanstih

Meaning: The part and the whole

Explanation

This sutra explains the relationship between a part and the whole. It is useful in fractions, percentages, and statistics.

Example

$$\text{If total students} = 200$$

$$\text{Girls} = 80$$

$$\text{Fraction} = 80/200 = 2/5$$

12. Sesanyan

Meaning: The remainder

Explanation

This sutra is used in division problems to quickly find the remainder and verify division answers.

Example

$$29 \div 6$$

$$\text{Quotient} = 4$$

$$\text{Remainder} = 5$$

13. Gunitasamuchyah

Meaning: Product of the sum

Explanation

This sutra is used to check multiplication answers using digit sums.

Example

$$32 \times 14 = 448$$

Digit sum:

$$32 \rightarrow 3+2 = 5$$

$$14 \rightarrow 1+4 = 5$$

$$5 \times 5 = 25 \rightarrow 2+5 = 7$$

$$448 \rightarrow 4+4+8 = 16 \rightarrow 1+6 = 7 \checkmark$$

14. Vistarana

Meaning: Expansion

Explanation

This sutra is used to expand algebraic expressions using distributive property.

Example

$$\begin{aligned}
 &(x + 4)(x + 6) \\
 &= x^2 + 6x + 4x + 24 \\
 &= x^2 + 10x + 24
 \end{aligned}$$

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15. Rupan

Meaning: Change of form

Explanation

This sutra converts numbers into easier forms such as fractions into decimals or vice versa.

Example

$$\begin{aligned}
 1/4 &= 0.25 \\
 \text{So } 3/4 &= 0.75
 \end{aligned}$$

16. Chidana

Meaning: By splitting

Explanation

This sutra breaks a number into parts to simplify multiplication.

Example

$$\begin{aligned}
 18 \times 7 \\
 &= (10 + 8) \times 7 \\
 &= 70 + 56 \\
 &= 126
 \end{aligned}$$

17. Paravartya Yojayet

Meaning: Transpose and apply

Explanation

This sutra is mainly used for fast division by numbers like 9, 99, 999 etc.

Example

Divide 458 by 9

Digit sum = $4+5+8 = 17$

Remainder = 8

18. Vilokanam

Meaning: By observation

Explanation

This sutra encourages pattern recognition to solve problems mentally.

Example

$$11 \times 12 = 132$$

$$11 \times 13 = 143$$

$$11 \times 14 = 154$$

19. Antyayoreva

Meaning: Only the last digits

Explanation

Used when last digits determine the final result, especially in squaring numbers ending with 5.

Example

$$45^2 = 2025$$



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20. Lopan-asthapan-abhyam

Meaning: Elimination and retention

Explanation

This sutra helps in simplifying algebraic expressions by eliminating common factors.

Example

$$x^2 + 7x + 10 = (x + 5)(x + 2)$$

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